



H1 Biology

Paper 1 Multiple Choice

8876/01

22 September 2021

60 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use paper clips, glue or correction fluid.

Write your name, civics group and registration number on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

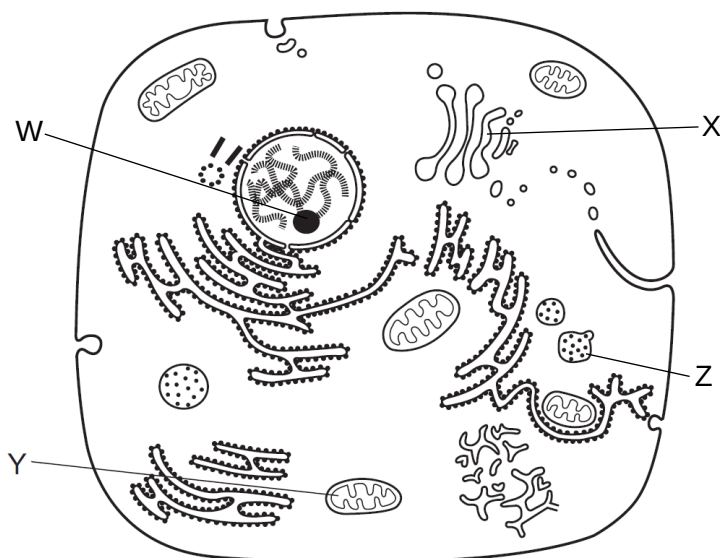
The use of an approved scientific calculator is expected, where appropriate.

This document consists of **17** printed pages and **3** blank pages.

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Answer **all** questions.

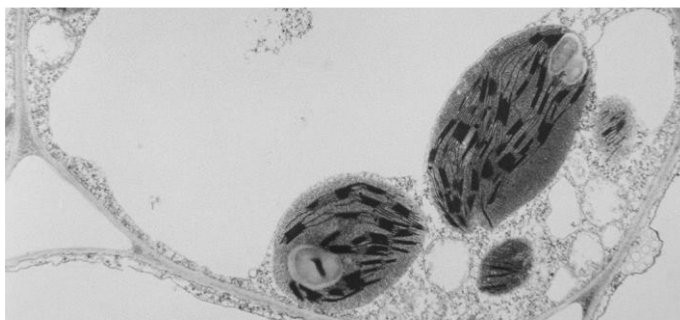
- 1 The diagram below shows a eukaryotic cell with four organelles labelled W to Z.



Which of the following rows correctly matches the organelle to its function?

	W	X	Y	Z
A	site of assembly of rRNA with proteins	glycosylation of cell surface receptors	oxidation of fatty acids and glucose	transport of proteins to be secreted
B	site of active gene transcription	synthesis and storage of lipids	site of Krebs cycle reactions	synthesis of cytosolic proteins
C	site of attachment of kinetochore proteins	sorting of proteins into vesicles	site of Calvin cycle reactions	transport of proteins to be modified
D	site of synthesis of ribosomal proteins	packaging of lysosomal enzymes	storage of glucose as glycogen	breakdown of foreign particles

- 2 The electronmicrograph below shows part of a eukaryotic cell.



Which of the following statements best explains why this is a plant cell?

- A Presence of a large central vacuole serves to regulate osmotic pressure to prevent lysis.
 - B** Stacks of thylakoids maximise surface area for proteins involved in photophosphorylation to be embedded in the membrane.
 - C Starch granules are clearly seen within the cytosol and serve as a compact energy storage.
 - D Presence of a cell wall determines which molecules can be taken up by the cell, and which will be repelled by the layer.
- 3 A student carried out the Benedict's test on three different solutions, X, Y and Z. All three solutions have the same concentration, but contain different carbohydrates.

A sample of each solution was incubated with acid for five minutes, and then heated in a boiling water-bath with Benedict's solution. The time taken for the first appearance of a colour change was recorded.

The results are shown in the table below.

solution	time for first appearance of a colour change with Benedict's solution / s
X	14
Y	300
Z	28

Which row correctly identifies the carbohydrate present in each solution?

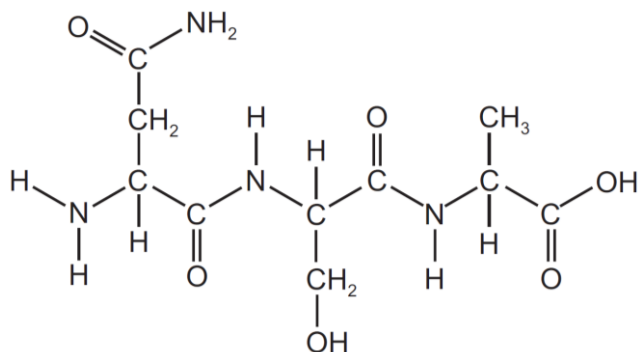
	X	Y	Z
A	glucose	starch	maltose
B	glucose	sucrose	lactose
C	lactose	starch	glucose
D	lactose	sucrose	maltose

- 4 Which row correctly matches the function with the structural features of cellulose, glycogen, collagen or triglyceride?

	molecules	functions	structural features		
			fibrous	molecules held together by hydrogen bonds	branched chains
A	cellulose triglyceride	mechanical support energy storage	✓ X	✓ X	X X
B	collagen cellulose	mechanical support mechanical support	✓ ✓	✓ X	X ✓
C	collagen glycogen	mechanical support energy storage	✓ X	✓ X	✓ ✓
D	glycogen triglyceride	mechanical support energy storage	X X	✓ X	✓ X

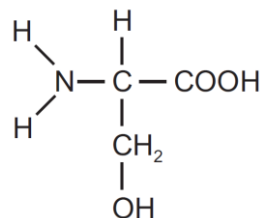
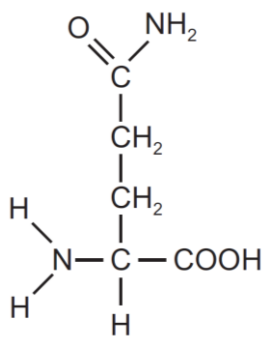
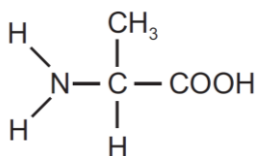
key: ✓ correct X not correct

- 5 The diagram shows the molecular structure of a tripeptide.

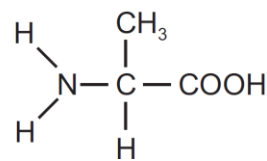
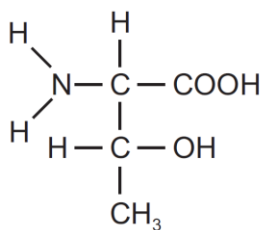
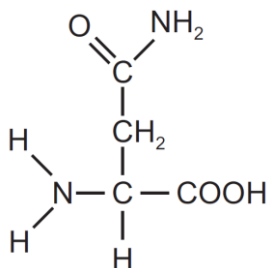


Which molecules would result from the complete hydrolysis of the tripeptide?

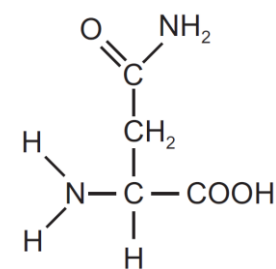
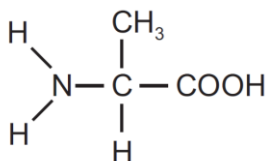
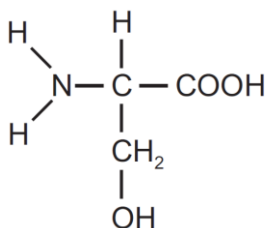
A



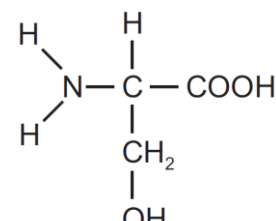
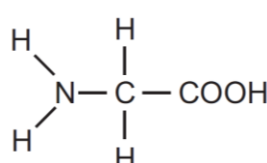
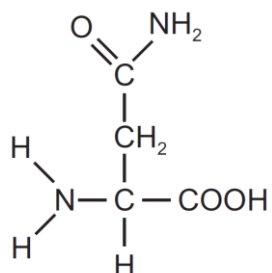
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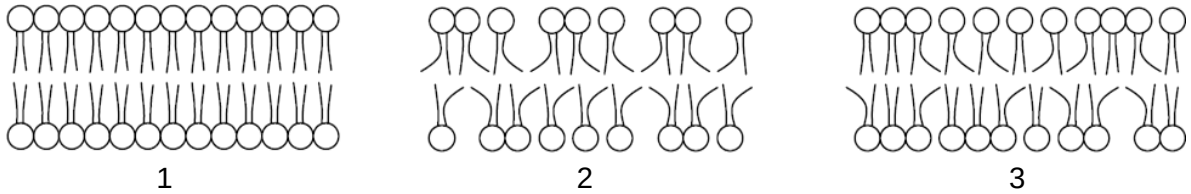
C



D



- 6 Cell membranes in three different organisms adapted for different climates are shown below.



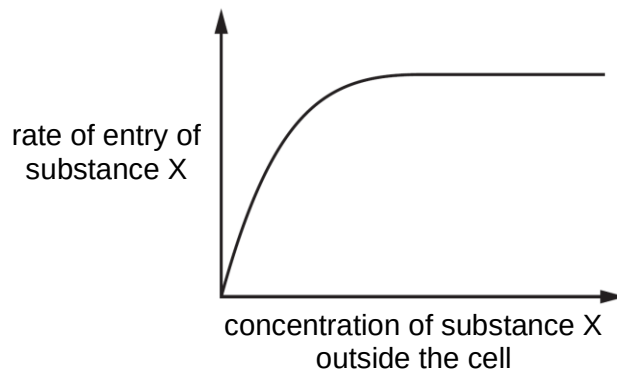
Several regions of the world are known to have the following climates:

- desert region is hot and dry
- tropical region is warm and wet
- temperate region is cool and moderate
- arctic region is cold and dry

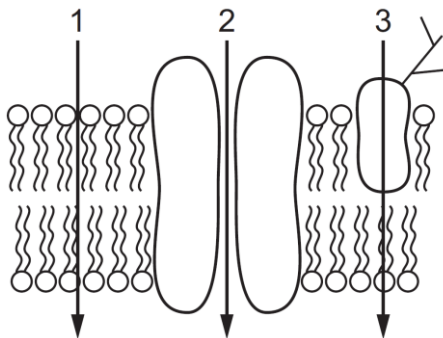
Which of the following correctly matches the regions to each type of cell membrane?

	1	2	3
A	desert	arctic	temperate
B	arctic	desert	tropical
C	desert	temperate	arctic
D	tropical	arctic	desert

- 7 The graph shows how the rate of entry of substance X into a cell changes as the concentration of substance X outside the cell increases.



The diagram below shows part of a cell surface membrane.

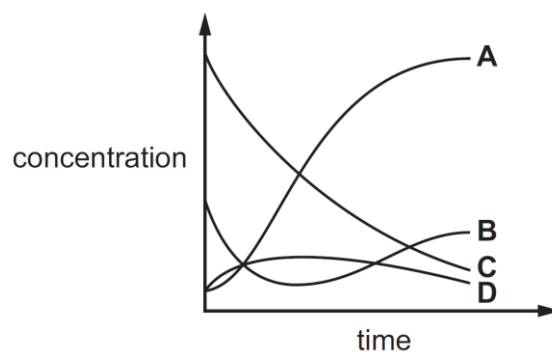


Based on the information given, which pathway(s) is/are likely for substance X to enter the cell?

- A** 1 only **B** 2 only **C** 1 and 2 **D** 1, 2 and 3

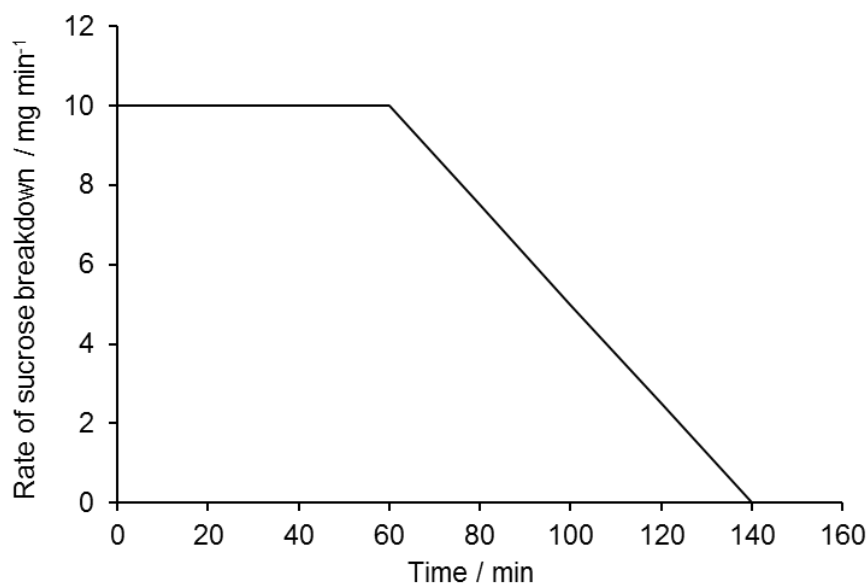
- 8 The graph below shows how the concentration of different components of an enzyme-catalysed reaction change with time.

Which line represents enzymes with empty active sites?



Answer = B

- 9 The graph shows the results of an investigation using invertase, an enzyme that breaks down sucrose into glucose and fructose. 1 g of sucrose was dissolved in 100 cm³ of water and 2 cm³ of a 1% invertase solution was added.



Which conclusion can be drawn from this information?

- A Between 0 and 60 min, the concentration of the substrate remains constant.
- B After 60 min, the concentration of enzymes becomes the limiting factor.
- C At 140 min, some of the enzyme molecules are denatured.
- D** Between 60 and 140 min, the concentration of the substrate is the limiting factor.

- 10** A number of blood transfusion laboratories around the world are hoping to use embryonic stem cells to produce large numbers of red blood cells.

Which statement correctly describes the differentiation of embryonic stem cells to red blood cells?

- A** Appropriate molecular signals are required to induce embryonic stem cells to differentiate into pluripotent blood stem cells and then into red blood cells.
- B** Appropriate molecular signals are required to induce totipotent embryonic stem cells to differentiate into blood stem cells and then into red blood cells.
- C** Appropriate chemical signals are required to induce embryonic stem cells to undergo differential gene expression to form red blood cells.
- D** Appropriate chemical signals are required to induce multipotent embryonic stem cells to undergo differential gene expression to form red blood cells.

- 11** Which of the following statement is false with regards to stem cells?

- A** Upon fertilisation, the zygote formed is a stem cell with the highest differentiation potential.
- B** Adult stem cells can be isolated from both infants and adults.
- C** Embryonic stem cells has a higher potency than adult stem cells as it can give rise to the placenta but somatic stem cells cannot.
- D** Both the specialised cells as well as stem cells from a living human can be used to create induced pluripotent stem cells.

- 12** A space probe brought back some living material from a distant planet. Proteins very similar to those found on Earth were present.

DNA of this living material, which exists as double helices, contains six different nucleic acid bases, identified as H, I, J, K, L and M. Among these, J, L and M were found to be pyrimidine bases.

Studies of the DNA from this living material gave the following results:

ratio of bases in double-stranded DNA	numerical value
H / J	1.00
(H + I) / (J + M)	1.02
(H + K) / (J + M)	1.14
(H + K) / (J + L)	1.01

Assuming that complementary base pairing only occurs between a purine and a pyrimidine, which of the following are valid conclusions from these data?

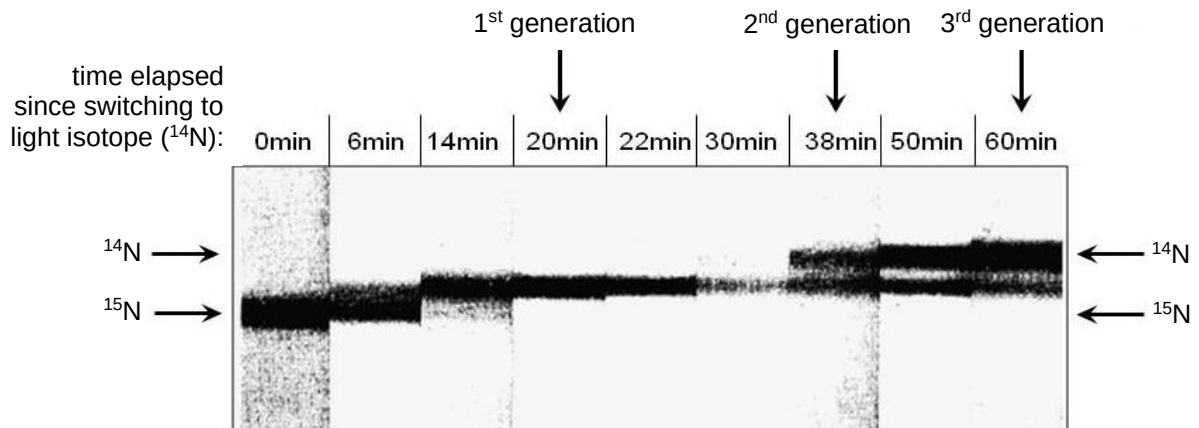
- 1 H base pairs with J.
- 2 H, I, K are likely to be purine bases.
- 3 It is probable that I base pairs with M, and K base pairs with L.

- A** 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 1, 2 and 3

- 13** In an investigation to study the mode of DNA replication, *Escherichia coli* (*E. coli*) cells were grown in a nutrient medium containing heavy isotope of nitrogen (^{15}N) for an extended period of time until all the DNA was labelled.

These *E. coli* cells were then transferred to a nutrient medium containing only light isotope of nitrogen (^{14}N) and were allowed to multiply over three generations. The DNA of the *E. coli* cells was then harvested at nine different time intervals.

Subsequently, density gradient centrifugation of these *E. coli* DNA using caesium chloride was performed.



Which statements are consistent with the results shown above?

- 1 In the 1st generation, only hybrid $^{14}\text{N}/^{15}\text{N}$ DNA is produced.
- 2 In the 2nd generation, there is no original parental DNA left.
- 3 In the 3rd generation, there is a 3:1 ratio of hybrid $^{14}\text{N}/^{15}\text{N}$ DNA to light $^{14}\text{N}/^{14}\text{N}$ DNA.
- 4 In the subsequent 4th generation, only light ^{14}N DNA strands will be synthesised.

- A** 1 and 2
B 1 and 4
C 2 and 3
D 3 and 4

- 14 The table below shows the anticodon sequences for some amino acids.

amino acid	anticodon (3' to 5')
alanine (A)	CGG / CGU
cysteine (C)	ACG
phenylalanine (F)	AAG
threonine (T)	UGU / UGA

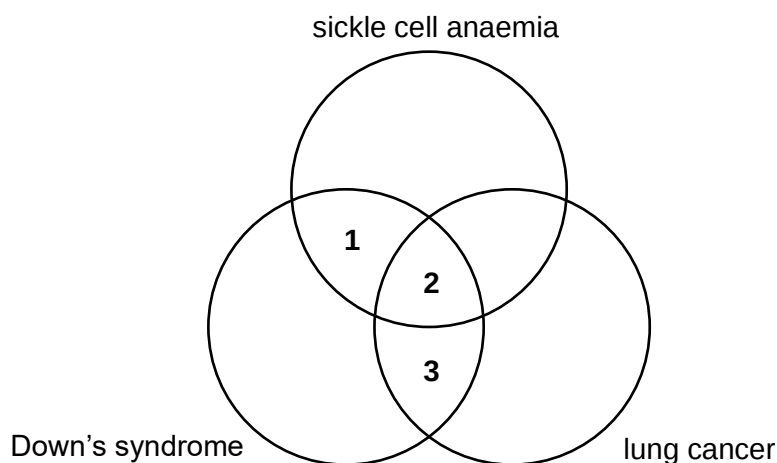
Part of a polypeptide chain has the sequence shown below.



Which of the following shows the correct sequence of nucleotides in the template DNA strand that codes for this section of the polypeptide chain?

- A** 5' – AGT GCA GGC GAA TGC – 3'
B 5' – CGT AAG CGG ACG TGA – 3'
C 5' – TGA ACG CGT AAG CGG – 3'
D 5' – TGC GAA GGC GCA AGT – 3'

- 15 Features of three diseases are identified with a number in the diagram below.



Which combination correctly describes these features?

	1	2	3
A	has a hereditary genetic component	may be caused by chromosomal translocation	can arise due to environmental causes
B	only arise due to genetic abnormalities	has a hereditary genetic component	may be caused by chromosomal translocation
C	can arise due to environmental causes	only arise due to genetic abnormalities	may be caused by chromosomal translocation
D	only arise due to genetic abnormalities	has a hereditary genetic component	can arise due to environmental causes

16 Which of the following statements about sickle cell anaemia are true?

- 1 Substitution of glutamate with valine has no effect on haemoglobin structure.
- 2 Polymerisation of HbS results in formation of fibrous precipitates in red blood cells.
- 3 A single copy of the β -globin gene must be mutated to result in sickle cell anaemia.
- 4 Elasticity of red blood cells is lost and they are unable to squeeze through capillaries.
- 5 Sickling of red blood cells can occur at high altitude due to low oxygen concentration.

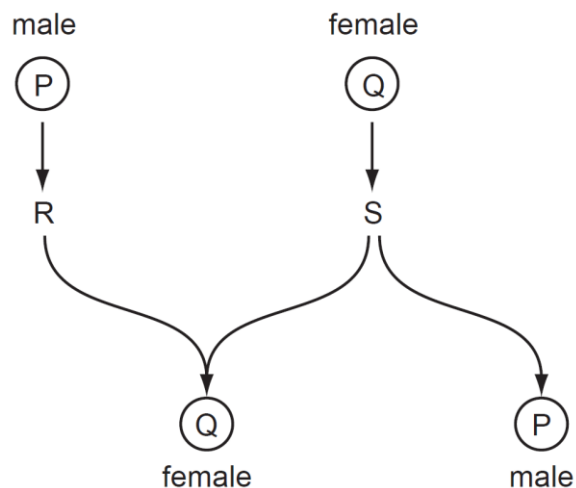
- A 1, 3 and 4
 B 2, 3 and 5
C 2, 4 and 5
 D 1, 2, 3, 4 and 5

17 During telophase of mitosis, a scientist stains the chromosomes of a diploid animal cell with a fluorescent dye to observe the telomeres. This cell has a diploid number of 22.

How many telomeres will the scientist observe in one of the nuclei?

- A 11 B 22 **C** 44 D 88

18 Sex determination in some insects such as bees and wasps is not controlled by sex chromosomes, but by ploidy level. Males are haploid, while females are diploid.



Which row correctly describes P, Q, R and S?

	P	Q	R	S
A	n	n	mitosis	mitosis
B	n	2n	mitosis	meiosis
C	2n	n	meiosis	meiosis
D	2n	2n	meiosis	mitosis

- 19 In *Drosophila*, vestigial wings and ebony body colour are recessive to the wild type and are not linked. Crossing two flies heterozygous for both characters produced 256 offspring.

How many offspring would be expected to have normal wings with ebony body colour?

- A 144 B 96 **C** 48 D 16

- 20 A cross between a round-leafed, tall plant and round-leafed, dwarf plant produced the following offspring:

121 round-leafed, tall plant
124 round-leafed, dwarf plant
42 oval-leafed, tall plant
37 oval-leafed, dwarf plant

Key

R – round leaf
r – oval leaf
T – tall
t – dwarf

What were the genotypes of the parents?

- A** RrTt x Rrtt
B RrTt x RRtt
C RrTT x Rrtt
D RrTT x RRtt

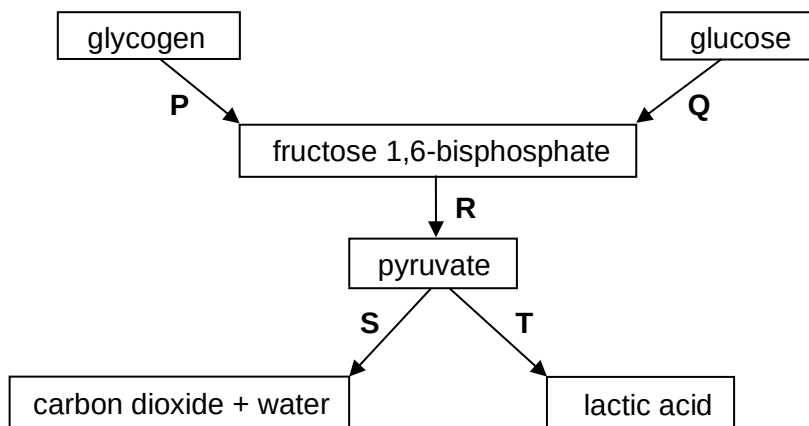
- 21 Huntington's disease is a genetic disorder that results in the degeneration of the nervous system. Several discoveries were made regarding Huntington's disease.

- The male offspring of affected males and all unaffected females have a 50% chance of getting the disease.
- In affected individuals, level of a protein necessary for brain cell development is approximately halved.
- The mutant allele codes for a non-functional protein as its DNA sequence contains tandem repeats within the coding region of the gene.
- Typically, symptoms do not start until after the age of 30 and the disease progressively worsens with age.
- People affected by the disease die during the age of 40 to 50.

What do these discoveries suggest about Huntington's disease?

- A The allele resulting in Huntington's disease is located on the X chromosome.
B Normal brain cell development requires expression of two normal alleles to produce normal levels of the critical protein and thus the disease-causing allele is dominant.
C It is controlled by multiple genes and symptoms can be influenced by environmental factors.
D The expression of the disease-causing allele is masked by the presence of a normal allele.

22 With reference to the diagram, relate processes P, Q, R, S, T to statements I, II and III.



- I NAD^+ is regenerated without the use of the electron transport system.
- II ATP is synthesised via substrate-level phosphorylation.
- III It can take place under anaerobic conditions.

	I	II	III
A	R	S	T
B	T	R	P
C	none of the above	S	R
D	S	Q	none of the above

23 Which statement about anaerobic respiration is correct?

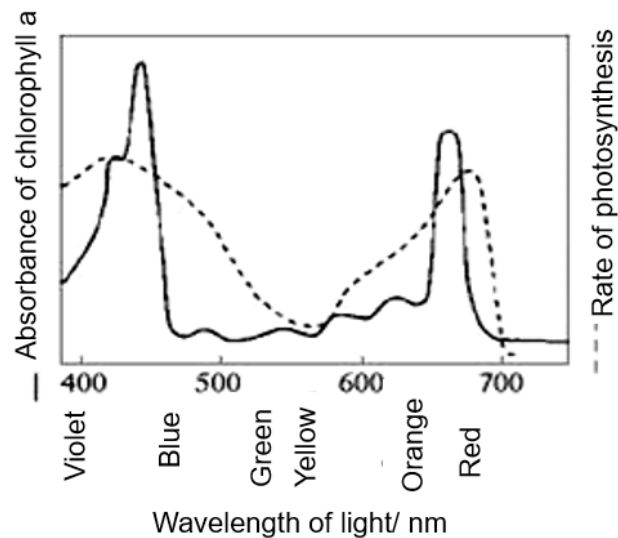
- A From one molecule of glucose, ethanol and lactate production yield the same amounts of ATP.
- B From one molecule of glucose, ethanol production yields less energy than lactate production.
- C From one molecule of glucose, ethanol production yields more energy than lactate production.
- D Yeast is able to respire ethanol for the production of ATP.

24 During aerobic respiration, ATP can be formed by oxidative phosphorylation in the electron transport system and by glycolysis.

In the complete oxidation of one molecule of glucose, approximately what percentage of ATP is formed by oxidative phosphorylation?

- A 90%
- B 70%
- C 30%
- D 15%

- 25 The action spectrum and absorption spectrum of photosynthetic pigments are similar because
- A the amount of energy absorbed by the pigments is the activation energy needed for photosynthesis.
 - B only certain wavelengths of light provide enough energy to make ATP during the light reaction.
 - C photosynthesis only occurs when the whole spectrum of light is absorbed.
 - D** the wavelengths of light absorbed by the pigments correlate with the rate of photosynthesis.
- 26 The graph shows the absorption spectrum for chlorophyll a and the action spectrum of a plant.

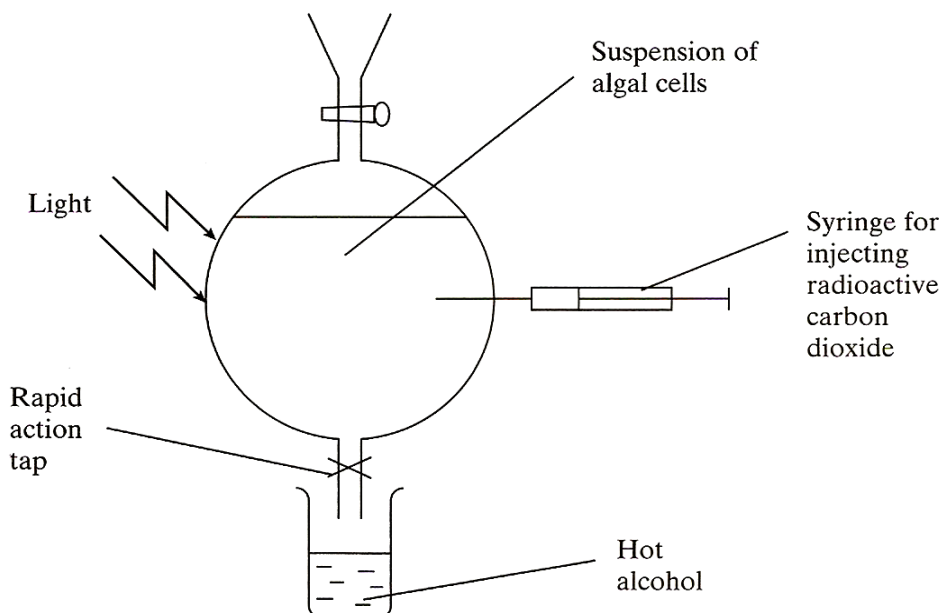


What is the reason for the difference in the absorption spectrum and the action spectrum?

- A Chlorophyll a is not present in the plant.
- B Chlorophyll a is the main pigment responsible for photosynthesis in the plant.
- C** There are other pigments responsible for photosynthesis in the plant.
- D Wavelengths of 480-630nm provide the least energy for photosynthesis.

- 27** An investigation was carried out to find out the sequence of biochemical changes that occur during photosynthesis. Radioactive carbon dioxide was added to a suspension of algal cells, and they were exposed to light. At intervals, samples of the suspension were removed into hot alcohol.

The experimental set up is shown below.



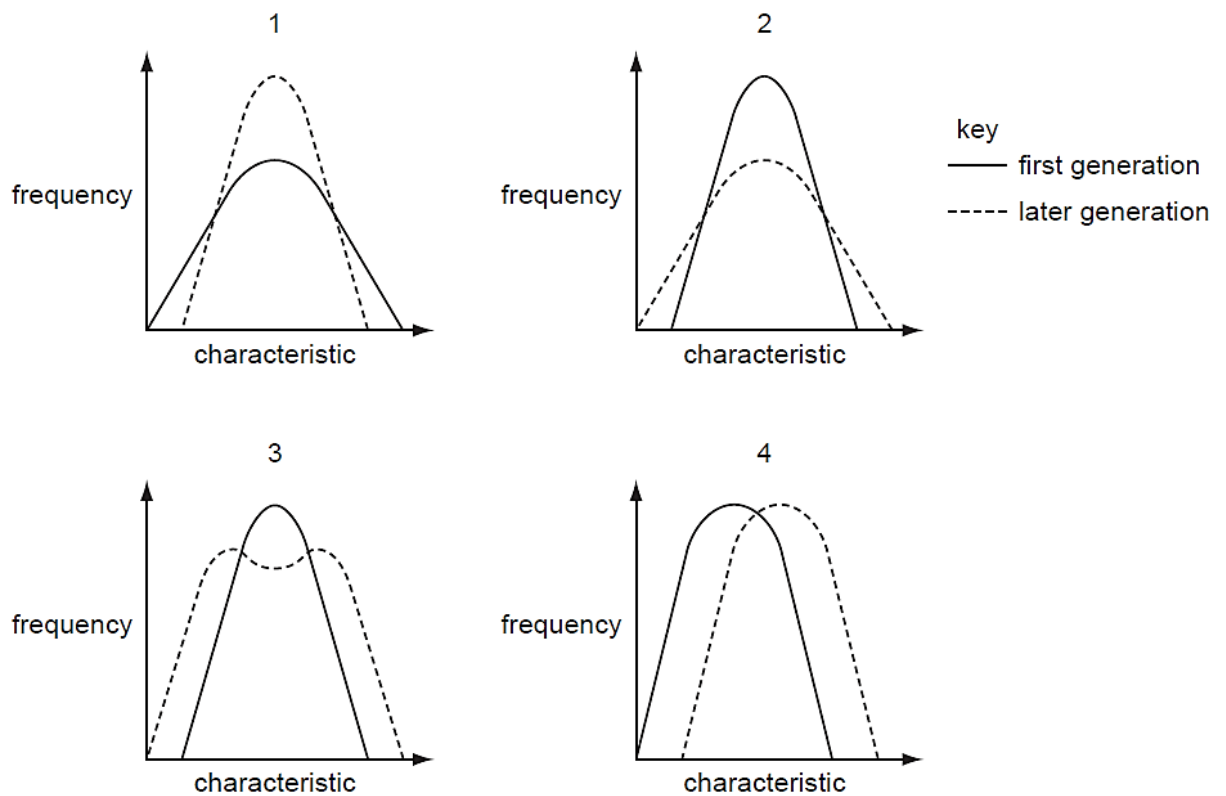
Samples were removed from the suspension at five different times, between 5 seconds and 600 seconds after the start of the experiment. These extracted samples were analysed for different radioactively labelled compounds. In each sample, the amount of radioactivity present in four different organic compounds, P, Q, R and S was measured. The table shows the results.

organic compound	amount of radioactivity present / arbitrary units				
	5 s	15 s	60 s	180 s	600 s
P	0.01	0.02	0.08	0.17	0.67
Q	1.00	2.00	3.10	3.15	3.15
R	0.10	1.50	2.20	2.30	2.40
S	0.05	0.11	0.16	1.00	1.00

What are the most likely identities of the four organic compounds?

	P	Q	R	S
A	G3P	GP	6 carbon intermediate	RuBP
B	GP	RuBP	G3P	6 carbon intermediate
C	RuBP	6 carbon intermediate	GP	G3P
D	RuBP	GP	G3P	6 carbon intermediate

- 28 The graphs show frequency against a measured characteristic in the first and later generation of an organism.



Which graph represents each type of natural selection?

	directional	disruptive	stabilising
A	1	2	3
B	2	3	4
C	3	1	2
D	4	3	1

- 29 A student recorded the shell colour and banding pattern of all empty shells of the brown-lipped snail, *Cepaea nemoralis*, found in the woods. Wide variations in shell colour and banding pattern, which are genetically controlled, was shown.

Gene **C** codes for the colour of the shell. There are six alleles.

C^B	C^{DP}	C^{PP}	C^{FP}	C^{DY}	C^{PY}
brown	dark pink	pale pink	faint pink	dark yellow	pale yellow
most dominant					least dominant

Gene **B** codes for the presence and absence of bands. There are two alleles.

B^O = unbanded, dominant allele

B^B = banded, recessive allele

Which conclusion drawn from this information is valid?

- A In the woods, brown, unbanded phenotypes will occur in the highest frequency and banded, pale yellow phenotypes will occur in the lowest frequency.
 - B** More than one selection pressure is likely to be acting to maintain the polymorphism shown in shell colour and banding pattern of the snails.
 - C Predation by birds, such as song thrush, is the only selection pressure acting on *C. nemoralis* in the woods.
 - D The proportion of each phenotype in the population of *C. nemoralis* in the woods will be similar to the proportion of each phenotype recorded for the empty shells.
- 30 The population size of northern elephant seals was reduced to 20 individuals in the 1890s due to overhunting. The population of northern elephant seals have since then rebounded to more than 30000.

Which of the following statement is true with regards to the northern elephant seals?

- A The reduction in population size to 20 northern elephant seals is an example of natural selection.
- B** Overhunting has led to a reduced gene pool in the population of northern elephant seals.
- C Genetic variation has recovered with the population size increasing to more than 30000.
- D The current population of northern elephant seals is less susceptible to genetic diseases.

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